Evidence at What Cost? Decoupling and Scaling in LLM-Based Document Relation Extraction

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Abstract

Large Language Model (LLM) based Document-level Relation Extraction (DocRE) now matches or exceeds encoder baselines in extraction accuracy, yet these methods output bare relation triplets without identifying which sentences support each prediction. Deployment contexts such as clinical knowledge bases and legal due-diligence systems require verifiable evidence that users can trace to source text. Encoder-based evidence methods (SAIS, DREEAM) provide supporting sentences but trail LLM approaches in accuracy, leaving a gap between extraction performance and output verifiability.

We formalize evidence-grounded DocRE as a structured generation task and compare Group Relative Policy Optimization (GRPO), Direct Preference Optimization (DPO), and Rejection Sampling Fine-Tuning (RSFT) across three fully evaluated scales (1.7B, 4B, 8B) with 14B validation, two model families, and two datasets. The naïve expectation is that evidence generation always costs extraction accuracy. Our experiments reveal that this cost is not fixed but forms a landscape shaped by capacity, paradigm, and data quality. Within the Qwen3 family, evidence is beneficial at 1.7B (-5.63pp Ign-F1) but approximately neutral at $\geq$4B (+0.25$\pm$0.47pp at 4B; positive = cost). Gradient-based optimization amplifies evidence-extraction competition (GRPO catastrophic, DPO gradual), while RSFT with Natural Language Inference (NLI) based scoring avoids decoupling entirely through output filtering rather than gradient pressure.

RSFT compresses training seed variance 4.6–31$\times$ across Qwen3 and LLaMA (complete pairwise seed separation, Cohen's d=2.58). Our 1.7B RSFT model (51.19 Ign-F1, std 0.09pp) approaches 8B Supervised Fine-Tuning (SFT) performance (50.19, std 3.23pp) at 4.7$\times$ fewer parameters. Test-set evaluation preserves model ranking across all evaluated scales (Spearman ρ=1.0).

1 Introduction

Document-level relation extraction (DocRE) requires identifying semantic relations between entity pairs across multiple sentences, a task where encoder-based methods such as Zhou et al. (2021) long dominated. Recent LLM-based approaches have closed this gap: Xue et al. (2024) fine-tune LLMs with structured decomposition, Li et al. (2024) frame it as multi-choice reading comprehension, and Xu et al. (2025) combine retrieval-augmented generation with structured knowledge. These methods approach or surpass encoder baselines on DocRED (Yao et al., 2019) and Re-DocRED (Tan et al., 2022) in extraction accuracy, yet every one outputs bare relation triplets (e_h, r, e_t) without indicating *which sentences* support each prediction. In deployment contexts—clinical knowledge bases, legal due-diligence pipelines, financial compliance systems—users cannot verify a prediction without manually locating its textual basis across a multi-page document.

Encoder-based methods have partially addressed this need: Xiao et al. (2022) introduce sentence-level evidence supervision as an auxiliary signal, and Ma et al. (2023) extend this with dynamic evidence retrieval through memory attention. Both produce evidence and improve extraction, but encoder methods now trail LLM-based approaches in accuracy. The community faces a gap: the most accurate extractors provide no evidence, while evidence-capable methods are no longer competitive.

This gap reflects a structural difference. In encoder methods, evidence retrieval is classification that regularizes the encoder at negligible cost; in LLM-based DocRE, evidence requires the model to *generate* additional output tokens per triple, increasing output length and decoding complexity. The naïve expectation is that this generation cost

always harms extraction accuracy. Our experiments show this expectation is wrong. The evidence tax is not a fixed penalty but a landscape shaped by model capacity, optimization paradigm, and training data quality. Within the Qwen3 family, evidence generation *benefits* extraction at 1.7B (-5.63pp Ign-F1, partly through halving output duplication) but is approximately neutral at $\geq$4B ($+0.25 \pm 0.47$pp at 4B). Gradient-based optimization amplifies this competition—GRPO collapses extraction under constrained precision, DPO progressively trades extraction for evidence—while discrete sample selection (RSFT) avoids the trade-off entirely by filtering outputs rather than pushing gradients.

We formalize evidence-grounded DocRE as a structured generation task and compare three post-training paradigms—GRPO (Shao et al., 2024), DPO (Rafailov et al., 2023), and RSFT (building on STaR; Zelikman et al. 2022)—across three fully evaluated scales (1.7B, 4B, 8B) with 14B validation, two model families, and two datasets. Our contributions:

1. **Evidence generation is not uniformly harmful.** We observe a scale-dependent evidence tax with a within-family capacity threshold: beneficial at 1.7B (-5.63pp) but approximately neutral at $\geq$4B ($+0.25 \pm 0.47$pp at 4B), challenging the assumption that evidence always costs extraction accuracy.

2. **Gradient pressure causes evidence-extraction decoupling.** We characterize a decoupling spectrum (GRPO catastrophic $\rightarrow$ DPO gradual $\rightarrow$ RSFT none) and introduce the Evidence-Decoupling Ratio (EDCR) as a diagnostic, validated at per-relation level ($\rho = -0.79$, $n = 76$, $p < 0.001$).

3. **RSFT compresses training variance 4.6–31$\times$.** NLI-based sample selection improves mean Ign-F1 across tested scales (+1.2 to +6.6pp); our 1.7B RSFT model (51.19 Ign-F1, std 0.09pp) approaches 8B SFT performance (50.19, std 3.23pp) at 4.7$\times$ fewer parameters, enabling deployment-efficient evidence-grounded extraction.

2 Related Work

Document-level relation extraction (DocRE) requires reasoning across sentence boundaries to identify relations between entity pairs. DocRED (Yao et al., 2019) established the first large-scale benchmark for this task, catalyzing a line of encoder-based methods that treat DocRE as classification over entity pair representations. ATLOP (Zhou et al., 2021) introduced adaptive thresholding for multi-label prediction and localized context pooling for entity representations. A parallel thread incorporates sentence-level evidence into these architectures. SAIS (Xiao et al., 2022) introduced evidence supervision as an auxiliary training signal and showed that joint evidence prediction improves extraction through data augmentation and ensemble inference. DREEAM (Ma et al., 2023) extended this approach with dynamic memory-based evidence retrieval that guides attention to supporting sentences. EIDER (Xie et al., 2022) jointly trains an evidence extractor and fuses evidence representations into the relation classifier. In all three systems, evidence identification is binary sentence classification over a fixed candidate set, a formulation that adds no output complexity. Recent work has recast DocRE for large language models. AutoRE (Xue et al., 2024) fine-tunes LLMs with QLoRA using a relation-head-facts decomposition; Li et al. (2024) frame extraction as multi-choice reading comprehension; RelPrior (Pi et al., 2025) uses relation priors to guide LLM-based extraction; GLiDRE (Armingaud and Besançon, 2025) presents a generalist lightweight approach; Ru-LLM (Zhang et al., 2025) incorporates logical rule constraints; and Xu et al. (2025) combine retrieval-augmented generation with structured knowledge. These methods match or surpass encoder baselines but produce bare triplets without evidence. The critical distinction lies in computational nature: in SAIS and DREEAM, evidence retrieval is classification that regularizes the encoder without increasing output length; in LLM-based DocRE, evidence requires additional output tokens per triple, imposing costs that depend on model capacity and optimization strategy. Whether this cost is inevitable or can be mitigated is the central question of our work.

Group Relative Policy Optimization (GRPO; Shao et al. 2024) eliminates the separate value network by computing advantages within groups. Subsequent work addresses failure modes: DAPO (Yu et al., 2025) prevents entropy collapse, Dr. GRPO (Liu et al., 2025) corrects length bias, and multi-objective variants (Liu et al., 2026; Ichihara et al., 2025) handle competing reward signals. Applications to structured IE include R1-RE (Dai et al.,

2025) and REAR (Luo et al., 2025). Direct Preference Optimization (Rafailov et al., 2023) reformulates RLHF as implicit reward maximization. Our decoupling spectrum is distinct from catastrophic forgetting in RL alignment (Jin et al., 2025), where models lose pre-trained capabilities across tasks; in our setting, extraction and evidence compete *within* a single task, resembling negative transfer in multi-task learning (Ruder, 2017; Standley et al., 2020; Yu et al., 2020), amplified by QLoRA's constrained parameter space (Gao et al., 2023; Moskovitz et al., 2024). RSFT avoids this intra-task competition by operating on discrete outputs rather than gradients, extending the sample-and-select principle (Wang et al., 2023) to structured IE with joint quality scoring.

Evidence faithfulness evaluation has received increasing attention: Li et al. (2025) propose dual-reward inference for faithful rationale generation, and Zhang et al. (2024) introduce fine-grained NLI-based faithfulness metrics for summarization. Our CED scoring builds on similar NLI-based principles but targets evidence-relation entailment rather than free-form rationales.

Zelikman et al. (2022) introduced STaR for iterative self-improvement; Dong et al. (2023) generalize this with reward-model-based sample filtering; recent analysis shows rejection sampling implicitly performs PPO-like optimization (Anil et al., 2025). Our RSFT is a single-round variant applied to evidence-grounded DocRE with CED/NLI scoring as a domain-specific quality filter. Because RSFT selects complete outputs rather than differentiating through a reward signal, it cannot exploit the extraction-evidence trade-off that gradient methods discover, inherently preserving joint quality.

3 Method

3.1 Task: Evidence-Grounded Document-Level Relation Extraction

Given a document $D = \{s_1, \ldots, s_n\}$ consisting of n sentences and a set of entities $\mathcal{E}$ identified within D, the task of evidence-grounded document-level relation extraction (DocRE) requires predicting, for each entity pair $(e_h, e_t) \in \mathcal{E} \times \mathcal{E}$, a relation $r \in \mathcal{R} \cup \{\texttt{NA}\}$ together with a set of supporting evidence sentences $E \subseteq D$. Standard DocRE predicts only the triple (e_h, r, e_t); the evidence-grounded formulation also requires E, which grounds each prediction in specific textual support.

In our LLM-based approach, the model generates structured outputs that jointly encode both components. For each predicted relation, the output follows the format `(head, relation, tail) | Evidence: [Si][Sj]`, requiring the model to produce complete evidence references inline with each triple rather than as a separate retrieval step. This design contrasts with encoder-based evidence methods such as SAIS (Xiao et al., 2022) and DREEAM (Ma et al., 2023), which frame evidence identification as sentence-level binary classification. Because classification over a fixed sentence set does not increase output length, these methods incur no generation overhead. Our LLM formulation requires additional tokens for each evidence reference, creating a generation burden that scales with the number of predicted triples and their evidence sentences. This scale-dependent cost is the central phenomenon we investigate.

3.2 Supervised Fine-Tuning Baseline

We fine-tune all models with LoRA (Hu et al., 2022) at rank r=64 and scaling factor α=128, applied to all linear projections (attention and feed-forward). The training data consists of DocRED (Yao et al., 2019) training documents converted to the structured output format described above, where each example maps a document and its entity set to the complete set of relation triples with evidence annotations. We also train a *no-evidence* variant under identical configuration but with evidence references removed from the output format, producing only `(head, relation, tail)`. This variant serves as an upper bound on extraction performance by eliminating the generation cost of evidence, and allows direct measurement of the evidence tax at each model scale. Detailed hyperparameters (learning rate, batch size, number of epochs) are reported in the Appendix.

3.3 Optimization Paradigms

We evaluate three post-SFT optimization paradigms that differ in how they incorporate quality signals from the CED scoring function (§3.4).

GRPO. Group Relative Policy Optimization (Shao et al., 2024) is an online RL method that eliminates the need for a separate value network by computing advantages within groups of sampled outputs. For each training document, the policy generates G candidate responses, which are scored by CED. The advantage of each response is

computed relative to the group mean, and the policy is updated to increase the probability of above-average outputs. We test two configurations: QLoRA with $G{=}4$ groups under 4-bit quantization, and bf16 full-precision LoRA with $G{=}8$. Because GRPO applies gradient updates that directly optimize the reward signal, evidence quality and extraction accuracy compete for gradient capacity. We analyze this multi-objective tension in Section 4.3.

DPO. Direct Preference Optimization (Rafailov et al., 2023) reformulates RLHF as a classification problem over preference pairs, without explicit reward models. We construct preference data from the SFT model's sampled outputs: for each document, the response with the highest CED score is labeled as preferred and the lowest-scoring response as rejected. DPO then adjusts the policy to increase the log-probability ratio between preferred and rejected completions. We train for 1 and 3 epochs to observe the temporal dynamics of evidence-extraction trade-offs under implicit optimization.

RSFT. Rejection Sampling Fine-Tuning is a single-round variant of iterative self-improvement methods such as STaR (Zelikman et al., 2022). The procedure is straightforward: the SFT model generates $N{=}8$ candidate outputs per training document, each candidate is scored by CED, and only the top-$k{=}1$ response is retained as training data. We use $k{=}1$ (best-of-N) selection throughout all experiments, selecting the single highest-scoring candidate per document for RSFT training data construction. This yields approximately 3,000 training examples per model (one per document; exact counts vary slightly by model due to generation failures). Unlike GRPO and DPO, RSFT does not apply gradient-based optimization pressure toward a reward signal. Instead, it improves training data quality through filtering: the model trains on its own best outputs rather than all outputs. This distinction is critical because discrete selection over complete outputs cannot exploit the extraction-evidence trade-off that gradient methods find. The contribution of RSFT in our setting is not the framework itself, which is well-established, but its application to evidence-grounded DocRE with CED as a domain-specific quality filter.

3.4 Contrastive Evidence Discrimination

CED is a scoring function that evaluates the faithfulness of predicted evidence sentences to their associated relation triples. It builds on Natural Language Inference (NLI) to measure whether evidence actually entails the predicted relation, rather than merely co-occurring with the relevant entities.

For a predicted triple (e_h, r, e_t) with evidence sentences E, CED constructs a positive NLI pair: the premise is the concatenation of sentences in E, and the hypothesis is a natural-language verbalization of the relation (e.g., "e_h has relation r with e_t"). An NLI model (`cross-encoder/nli-deberta-v3-base`) produces an entailment probability p^+ for this pair. The contrastive component introduces hard negatives drawn from the same document. For each triple, we collect sentences that mention e_h or e_t but are not in the predicted evidence set. These entity-sharing distractors serve as negative premises, and the NLI model scores each to obtain $p_1^-, \ldots, p_m^-$. The CED score for a single triple is:

$$\text{CED}(e_h, r, e_t, E) = p^+ - \max_j p_j^- \qquad (1)$$

This formulation forces the scoring function to distinguish sentences that genuinely support the relation from sentences that merely mention the same entities. A high CED score indicates that the evidence provides relation-specific support beyond entity co-occurrence.

The document-level CED score aggregates over all predicted triples, weighted by extraction correctness: triples that match ground-truth relations contribute their CED scores, while incorrect triples receive zero credit. This coupling ensures that CED rewards both faithful evidence and accurate extraction jointly.

We also evaluate a *flat-NLI* baseline that uses only the positive entailment score p^+ without the contrastive term. CED and flat-NLI produce comparable RSFT outcomes (mean Rel-F1 difference of 0.02pp across available seeds; Section 5), confirming that the contrastive formulation is not essential for data selection quality. We adopt CED as the default because its per-triple discrimination scores support interpretable error analysis.

3.5 Evidence-Decoupling Ratio

We introduce the Evidence-Decoupling Ratio (EDCR) as a diagnostic metric that quantifies the

degree of misalignment between a model's extraction and evidence capabilities. EDCR is not a training component; it serves purely as an analytical tool for comparing optimization methods.

EDCR is defined as:

$$\text{EDCR} = \frac{\text{Evi-Recall}_{\text{TP}}}{\text{Rel-Recall}} \tag{2}$$

where $\text{Evi-Recall}_{\text{TP}}$ is evidence sentence recall computed only over correctly predicted triples (true positives), and Rel-Recall is overall relation extraction recall. The numerator captures evidence quality on the subset of triples the model gets right; the denominator captures how many triples the model identifies in total.

The intuition is as follows. A model that correctly identifies a few triples and provides excellent evidence for them, but misses most triples entirely, will have high $\text{Evi-Recall}_{\text{TP}}$ and low Rel-Recall, yielding a high EDCR. This pattern signals that evidence quality has decoupled from extraction capability: the model produces faithful evidence only within a narrow comfort zone.

EDCR operates at two diagnostic levels. At the method level, comparing EDCR across seven independent method configurations reveals a negative correlation with Rel-F1 (Spearman $\rho{=}-0.89$, $n{=}7$, $p{=}0.007$). Because EDCR shares Rel-Recall with Rel-F1 by construction, this correlation should be interpreted as a diagnostic pattern rather than an independent discovery; its primary value lies in per-relation analysis. We validate EDCR's diagnostic utility through per-relation analysis (§5), which avoids the shared-component concern present in cross-method correlation. At the per-relation level, computing EDCR for individual relation types reveals a robust negative correlation with Rel-F1 ($\rho{=}-0.79$, $n{=}76$, $p{<}0.001$) that is independent of Evi-F1 ($\rho{=}-0.06$, $p{=}0.58$). This independence means EDCR captures relation-specific extraction-evidence misalignment that aggregate evidence metrics miss. The SFT baseline EDCR (0.6982) reflects the inherent cost of evidence generation; EDCR values substantially exceeding this level (e.g., GRPO QLoRA at 0.9876) indicate RL-induced decoupling beyond what evidence production alone explains, while values below the baseline (e.g., RSFT at 0.6503) reflect improved co-optimization.

4 Experiments

4.1 Experimental Setup

Dataset. We conduct experiments on DocRED (Yao et al., 2019), a large-scale document-level relation extraction benchmark containing 3,053 training and 998 development documents spanning 96 relation types. Each document requires identifying relational triples across multiple sentences and grounding each triple with supporting evidence sentences. We use Re-DocRED (Tan et al., 2022) for cross-dataset generalization evaluation.

Models. Our primary model is Qwen3-4B. To study scale effects, we also train Qwen3-1.7B and Qwen3-8B under identical configurations. We include LLaMA-3.1-8B as a cross-family validation point.

Training. All models use LoRA (Hu et al., 2022) with rank $r{=}64$ and $\alpha{=}128$ over a structured output format that jointly produces relation triples and evidence sentences. For RSFT, we sample $N{=}8$ candidate outputs per document and select the top-$k{=}1$ response ranked by CED scoring (§3.4), which combines NLI-based evidence faithfulness with extraction accuracy. GRPO experiments use group sizes $G{=}4$ (QLoRA, 4-bit) and $G{=}8$ (bf16 full-precision LoRA). DPO constructs preference pairs from CED-ranked outputs.

Metrics. **Ign-F1** (primary) ignores training-set triples per the official DocRED protocol. **Rel-F1** measures extraction F1 on all triples. **Evi-F1 (TP-only)** measures evidence F1 conditioned on correct triples; **Evi-F1 (joint)** follows the DREEAM protocol (Ma et al., 2023) for cross-method comparison. **EDCR** (§3.5) diagnoses extraction-evidence decoupling. All results use the development set.

4.2 Main Results

Table 1 presents the primary comparison on Qwen3-4B. The SFT baseline exhibits high seed sensitivity (mean Ign-F1 43.58, std 3.80pp, range 39.74–47.34). Removing evidence entirely (no-evidence SFT, s42) reaches 47.21 Ign-F1.

RSFT CED achieves 44.82 Ign-F1 (+1.24 over SFT mean) with substantially lower variance (std 0.51pp), though these reflect different sources of variation (Section 4.4). Even the weakest RSFT seed (44.23) exceeds the SFT mean. Evidence quality improves (Evi-F1 80.96 vs. 79.61) and coupling

Table 1: Main results on DocRED dev (Qwen3-4B). Ign-F1 shows mean ± std$^{(\text{seeds})}$; other metrics are same-seed means. Best **bold**; best evidence-generating underlined. Per-seed: Appendix 5.

Method	Ign-F1	Rel-F1	Evi-F1	EDCR
SFT baseline	43.58$_{\pm 3.80}^{(3)}$	44.39	79.61	0.6797
SFT no-evidence	**47.21**	**48.76**	–	–
RSFT CED	44.82$_{\pm 0.51}^{(3)}$	45.80	80.96	0.6487

tightens (EDCR 0.6487 vs. 0.6797). Per-seed results appear in Appendix B.

4.3 Evidence-Extraction Decoupling Spectrum

Because evidence and extraction share gradient capacity, different optimization paradigms handle this competition in qualitatively different ways. We organize the outcomes along a *decoupling spectrum* from catastrophic collapse to no decoupling, shown in Table 2 (Figure 1 in Appendix).

GRPO QLoRA: catastrophic collapse. Under 4-bit quantization (G=4), GRPO drives Rel-F1 to 1.38 while Evi-F1 rises to 85.04 (EDCR 0.9876): the model produces well-formed evidence but loses extraction capability. This extreme decoupling co-occurs with constrained precision; full-precision GRPO (below) avoids it.

GRPO bf16: mild, variable. Full-precision GRPO (G=8) avoids catastrophe, with the best checkpoint achieving Rel-F1 45.49, Evi-F1 80.86, and EDCR 0.6318 (below SFT baseline 0.6797). Both sub-tasks improve jointly, though the variable training trajectory requires careful checkpoint selection.

DPO: gradual decay. DPO progressively trades extraction for evidence: Rel-F1 drops from 40.34 (1 epoch) to 34.29 (3 epochs)[1] while evidence improves (Evi-F1 80.88→84.03, EDCR 0.7139→0.7294).

RSFT: no decoupling. RSFT maintains both capabilities simultaneously (Rel-F1 45.80, Evi-F1 80.96, EDCR 0.6487) because discrete output selection cannot exploit the extraction-evidence tradeoff that gradient methods find.

4.4 Scale-Dependent Evidence Tax

The evidence tax, defined as Ign-F1(no-evidence) − Ign-F1(with-evidence) on the development set, is scale-dependent. Under matched evaluation conditions (`max_new_tokens`=2048, format-compliant intersection; Appendix E), the training-time tax reveals a within-family capacity threshold: evidence generation benefits 1.7B (−5.63pp; Appendix E), where it also halves the output duplication rate (60.5%→33.0%; Appendix K), but is approximately neutral at ≥4B (+0.25±0.47pp at 4B, n=3; +0.04pp at 8B; +0.03pp at 14B). Table 3 presents RSFT performance and inference-time evidence tax across five models (Figure 2 in Appendix).

RSFT navigates this capacity landscape through quality-based data selection rather than gradient pressure. Mean Ign-F1 improves across tested scales: +6.55pp at 1.7B, +1.24pp at 4B, +4.14pp at 8B, and +3.79pp at 14B (Table 3). RSFT from an independent SFT base (s44, Method B) achieves 51.52 Ign-F1 at 4B (+4.18pp over that base). This result confirms that RSFT gains are not contingent on a specific SFT checkpoint.

Beyond mean performance, RSFT substantially improves training reliability. Three RSFT seeds from a shared SFT checkpoint yield standard deviations of 0.09pp at 1.7B, 0.51pp at 4B, and 0.50pp at 8B (n=3 each), compared to SFT cross-seed variation of 2.79pp, 3.80pp, and 3.23pp respectively. While these measure different sources of variance (SFT reflects full training runs from different random seeds; RSFT reflects additional fine-tuning from a fixed base), the practical consequence is direct: across all Qwen3 scales with ≥3 seeds, variance compression is 31× at 1.7B, 7.5× at 4B, and 6.5× at 8B. The 1.7B RSFT model (51.19 Ign-F1, std 0.09pp) approaches 8B SFT performance (50.19, std 3.23pp) at 4.7× fewer parameters—comparable in mean yet 36× more predictable.

Cross-family validation on LLaMA-3.1-8B confirms these patterns. All three RSFT seeds (55.87–

Table 2: Evidence-extraction decoupling across RL paradigms (Qwen3-4B). Severity indicates the degree of extraction collapse relative to evidence improvement.

Paradigm	Config	Rel-F1	Evi-F1	EDCR	Severity
GRPO	QLoRA G=4	1.38	85.04	0.9876	Catastrophic
GRPO	bf16 G=8	45.49	80.86	0.6318	Mild
DPO	bf16 1ep	40.34	80.88	0.7139	Gradual
DPO	bf16 3ep†	34.29	84.03	0.7294	Gradual
SFT	baseline‡	44.39	79.61	0.6797	None
RSFT	CED‡	45.80	80.96	0.6487	None

†Earlier evaluation protocol. ‡Three-seed mean. Severity reflects training trajectory stability (propensity for extraction collapse during optimization), not the final-checkpoint EDCR value.

[1] DPO 3-epoch results use an earlier evaluation protocol; included for trend analysis.

57.14) exceed all three SFT seeds (48.02–53.84), yielding complete pairwise seed separation (Cohen's d=2.58) and 4.6× variance compression (std 2.92pp → 0.64pp). The SFT seed lottery is starkest at s44 (48.02 Ign-F1, 24.4% truncation, elevated Evi-F1 85%), a degenerate outcome that RSFT eliminates entirely (s44: 55.87). This robustness extends to inference-time format changes: LLaMA SFT models evaluated without evidence at 1024 tokens suffer 38–42% truncation with Ign-F1 collapsing to ∼0.33 (s43/s44), while RSFT models truncate only 18–19% with stable performance (55.41±0.13pp), indicating that CED filtering selects for compact output structure.

For RSFT models, the inference-time evidence tax (same model evaluated with vs. without evidence output) preserves the benefit direction at 1.7B (−7.75pp, confirmed with 2048-token no-evidence evaluation that rules out truncation). At 4B, where the training-time tax is approximately neutral (+0.25±0.47pp; Appendix E), the RSFT inference-time tax is similarly small (+0.94pp). At 8B, the tax reverses to a benefit (−0.78pp), and at 14B the benefit is largest (−4.54pp). RSFT thus maintains or improves extraction quality with evidence across the Qwen3 family. LLaMA-3.1-8B RSFT shows a larger inference-time evidence benefit (−2.17±0.76pp, three-seed mean, vs. −0.78pp for Qwen3-8B), confirming that both performance gains and training reliability generalize across model families.

The effect of RSFT on inference-time evidence dependence is heterogeneous. At 1.7B, RSFT maintains strong evidence benefit (−5.43pp[‡]), comparable to SFT (−4.86pp). At 4B, RSFT introduces a mild evidence cost (+1.51pp), the only Qwen3 scale with a positive RSFT test tax. Within the Qwen3 family at ≥8B, RSFT approaches evidence neutrality, with test tax magnitudes below 1pp. Cross-architecture, LLaMA-3.1-8B shows partial compression: SFT tax of −4.94pp reduces to −2.65pp under RSFT, consistent with the cross-family variation observed in SFT evidence tax (Appendix E).

4.5 Evidence Strategy Comparison

We compare joint generation against post-hoc BM25 retrieval (entity-constrained, k=2), using Evi-F1 (joint) with the DREEAM-protocol denominator (Ma et al., 2023). Post-hoc retrieval achieves higher Rel-F1 (48.76) by avoiding the evidence tax but lower Evi-F1 (joint) of 34.7 (EDCR 0.766).

Table 3: RSFT performance and inference-time evidence tax across model scales (DocRED dev Ign-F1). LLaMA-3.1-8B included for cross-family validation. Mean ± std$^{(\text{seeds})}$.

Model	SFT w/ evi	RSFT	RSFT Tax[†]	RSFT Gain
Qwen3-1.7B	$44.64_{\pm 2.79}{}^{(3)}$	$51.19_{\pm 0.09}{}^{(3)}$	−7.75	+6.55
Qwen3-4B	$43.58_{\pm 3.80}{}^{(3)}$	$44.82_{\pm 0.51}{}^{(3)}$	+0.94	+1.24
Qwen3-8B	$50.19_{\pm 3.23}{}^{(3)}$	$54.33_{\pm 0.50}{}^{(3)}$	−0.78	+4.14
Qwen3-14B	55.07	58.86	−4.54	+3.79
LLaMA-3.1-8B	$51.09_{\pm 2.92}{}^{(3)}$	$56.54_{\pm 0.64}{}^{(3)}$	$-2.17_{\pm 0.76}{}^{(3)}$	+5.45

[†]Inference-time: same RSFT model with vs. without evidence output. No-evi eval uses `max_new_tokens`=2048; with-evi eval uses 1024 (1.7B uses 2048 for both). Tax = no-evi − with-evi; positive = cost. Seeds: 1.7B s42; 4B Method B s44; 8B, 14B: s42; LLaMA: three-seed mean. Training-time tax (matched@2048 fair evaluation) in Appendix E.

RSFT CED reaches 36.1 (EDCR 0.654), a 1.4-point evidence advantage. CED reranking at inference (best-of-N=8 by CED score) achieves Rel-F1 48.64 and Evi-F1 81.88 (TP-only), approaching no-evidence extraction quality while preserving evidence.

4.6 Generalization

Cross-architecture validation on LLaMA-3.1-8B (Table 3) shows the largest RSFT gain (+5.45pp) with complete pairwise seed separation (§4.4). The RSFT improvement transfers across datasets: on Re-DocRED (Tan et al., 2022), RSFT achieves 41.45 Ign-F1 (+4.86pp over SFT 36.59), matching the DocRED gain (+5.08pp). Our best joint Evi-F1 (36.1, DREEAM-protocol) falls below supervised DREEAM (54.01% (Ma et al., 2023)), a gap driven by extraction performance rather than evidence quality (TP-only Evi-F1 exceeds 80%). On the Re-DocRED test set, RSFT Ign-F1 drops approximately 20pp relative to dev across all evaluated scales (e.g., 14B: 58.9→38.5), attributable to 2.82× higher annotation density in test (34.7 vs. 12.3 gold triples per document); model ranking is perfectly preserved (Spearman ρ=1.0; Table 9 in Appendix). Full results appear in Appendix I.

5 Analysis

EDCR as a diagnostic. EDCR (§3.5) correlates with extraction quality at two levels: cross-method ($\rho = -0.89$, $n = 7$, $p = 0.007$; Table 2) and per-relation ($\rho = -0.79$, $n = 76$, $p < 0.001$), the latter providing validation independent of method-level sample size and orthogonal to Evi-F1 ($\rho = -0.06$, $p = 0.58$; Appendix L). RSFT avoids decoupling because discrete output selection cannot exploit the extraction-evidence trade-off (Appendix F). CED operates effectively at the response

level but fails at sentence-level granularity (Evi-F1 -7.7pp), as filtering individual sentences disrupts co-reference chains.

Ablation. CED and flat-NLI produce comparable RSFT performance (Rel-F1 45.63 vs. 45.61); stricter selection (k=1) outperforms k=3 by 1.02 Ign-F1 despite 2.96× fewer examples; CED filtering is insensitive to NLI scorer capacity (+1.10pp with DeBERTa-v3-large vs. v3-base, within seed noise; Appendix B).

Error patterns. RSFT reduces truncation FN, hallucinated FP, and duplicates by 6–22% across scales, but evidence quality is unchanged ($\Delta <$ 0.4pp; Appendix K). RSFT improves *what* the model extracts but not *how well* it grounds extractions.

Practical implications. Three guidelines for evidence-grounded IE: (1) avoid online RL for multi-objective generation where capability coupling matters; (2) prefer RSFT with strict selection (k=1) over training data quantity (4.6–31× variance compression across Qwen3 and LLaMA); (3) consider CED reranking at inference for deployment scenarios requiring both extraction accuracy and evidence quality.

6 Conclusion

We formalize evidence-grounded document-level relation extraction as a structured generation task and compare three post-training paradigms across three fully evaluated scales (1.7B, 4B, 8B) with 14B validation, two model families, and two datasets. The evidence tax is not a fixed penalty: it forms a landscape where capacity determines whether evidence regularizes or is absorbed, optimization paradigm determines decoupling severity, and data quality filtering enables RSFT to navigate without gradient-induced trade-offs.

RSFT compresses training variance by up to 31× while improving mean extraction across all tested scales. A 1.7B RSFT model approaches 8B SFT performance at 4.7× fewer parameters, demonstrating that evidence-grounded extraction can be both accurate and deployment-efficient.

Future work should investigate targeted evidence rewards—RSFT improves extraction but leaves evidence quality largely unchanged—and extend evaluation to non-Wikipedia domains where evidence annotation is currently unavailable.

7 Limitations

Primary development is on the DocRED development set; Re-DocRED test-set evaluation preserves model ranking (Spearman ρ=1.0) but shows a 18.9–21.9pp absolute drop attributable to higher annotation density (2.82× more gold triples per document). Exact-match F1 penalizes surface-form-different entity mentions (Jiang et al., 2024), likely underestimating absolute extraction capability of generative models, though relative comparisons between methods remain valid.

We evaluate on two datasets (DocRED, Re-DocRED), both derived from Wikipedia. Cross-domain evaluation is constrained by the absence of evidence-annotated DocRE datasets beyond Wikipedia; this limitation is shared by all prior evidence-aware DocRE work (SAIS, DREEAM, EIDER).

We evaluate one algorithm per RL family (GRPO, DPO) with single hyperparameter settings (KL coefficient 0.01, β 0.1); other algorithms or settings may produce different decoupling behaviors. CED scoring depends on an external NLI model, and the RSFT ablation uses limited seeds (four for CED, two for flat-NLI). Evidence tax patterns span two model families (Qwen3, LLaMA) but may not generalize to other architectures or scales beyond 14B.

EDCR correlates with extraction quality at both method level ($\rho=-0.89$, n=7) and per-relation level ($\rho=-0.79$, n=76), but has not been validated against human judgments of evidence-extraction alignment. Future work should assess whether EDCR rankings correspond to practitioner assessments of output quality.

RSFT improves extraction across scales but evidence quality is unchanged (Evi-F1 $\Delta <$ 0.4pp at all scales; Appendix K). CED-based selection filters for joint quality but does not specifically optimize evidence grounding, suggesting that targeted evidence rewards may be needed to improve evidence quality beyond what data selection achieves.

At 1.7B scale, 44.4% of outputs exhibit repetitive patterns (vs. negligible at 4B+); small models may require decoding-time interventions (e.g., repetition penalty) for reliable deployment.

8 Ethics Statement

This work uses publicly available benchmarks (DocRED, Re-DocRED) derived from Wikipedia, containing no private or sensitive information. All

models are trained on existing annotated data without human subjects involvement. Our evidence-grounding approach aims to improve the verifiability of relation extraction outputs, which supports responsible deployment in high-stakes domains such as clinical and legal information systems by enabling human verification of model predictions. We release code and model outputs to support reproducibility.

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A Training Hyperparameters

Table 4 lists the hyperparameters for all training configurations. All methods use LoRA with rank r=64 and α=128 applied to all linear projections (attention and feed-forward). SFT and RSFT share the same optimizer settings; RSFT differs only in data construction (top-1 CED-selected outputs from N=8 candidates). GRPO and DPO use lower learning rates to stabilize post-SFT optimization.

B Full Ablation Results

Table 5 reports the complete RSFT ablation across scoring functions, top-k values, and random seeds. RSFT CED k=1 achieves a four-seed mean Ign-F1 of 44.66 (standard deviation 0.49), confirming robust improvement over the SFT s42 baseline (39.74). The three-seed mean reported in Table 1 (44.82, seeds 42/43/44) is slightly higher because it excludes the lowest seed (s45: 44.16). Random selection RSFT k=1 (41.14 Ign-F1), which samples from the same candidate pool without CED scoring, improves only +1.40pp over SFT s42 versus +4.92pp for CED RSFT, confirming that CED scoring precision drives the majority of the RSFT gain. Increasing k to 3 slightly degrades performance (43.64 Ign-F1), consistent with the intuition that stricter selection yields higher-quality training data. The flat-NLI scoring variant produces comparable Rel-F1 (mean 45.61 vs. 45.63 for CED, a difference of 0.02pp). Both CED and flat-NLI improve substantially over the SFT baseline while maintaining or improving Evi-F1, confirming that the RSFT

Table 4: Training hyperparameters for all optimization paradigms. All experiments use Qwen3-4B as the base model with LoRA adaptation. A dash indicates that the parameter is not applicable to the given method.

Parameter	SFT	RSFT	GRPO QLoRA	GRPO bf16	DPO
Base model	Qwen3-4B	Qwen3-4B	Qwen3-4B	Qwen3-4B	Qwen3-4B
LoRA rank r	64	64	64	64	64
LoRA α	128	128	128	128	128
Learning rate	2e-5	2e-5	5e-6	5e-6	5e-7
Batch size (effective)	16	16	4	8	4
Epochs	3	3	–	–	1, 3*
Max steps	–	–	100	100	–
Gradient accumulation	4	4	16	4	4
Max new tokens	1024	1024	1024	1024	1024
Quantization	–	–	NF4 4-bit	bf16	bf16
Group size G	–	–	4	8	–
KL coefficient	–	–	0.01	0.01	–
β (DPO)	–	–	–	–	0.1
Candidates N	–	8	–	–	8
Top-k selection	–	1	–	–	–
Training samples	3,053	2,718	3,053	3,053	3,053
Scoring function	–	CED	CED	CED	CED

*DPO is trained for both 1 and 3 epochs to observe temporal dynamics of the evidence-extraction trade-off (Section 4.3).

gains are driven by quality-based data selection rather than by the specific contrastive formulation.

NLI scoring comparison. Across all available seeds, CED RSFT achieves mean Rel-F1 of 45.63 (seeds 42, 43, 44, 45: range 45.14–46.21), and flat-NLI RSFT achieves mean Rel-F1 of 45.61 (seeds 42, 43: range 45.35–45.86). The 0.02pp difference suggests comparable performance: CED's contrastive design does not substantially outperform simpler NLI scoring for RSFT data selection. We adopt CED as the default because its contrastive formulation produces per-triple discrimination scores that support interpretable error analysis.

Top-k selection. Best-of-8 (k=1) yields mean Ign-F1 44.66 and Rel-F1 45.63, while top-3-of-8 (k=3) produces 43.64 and 44.54. The selected training data confirms the quality gap: k=1 data has mean extraction F1 of 61.57 and mean CED of 0.6683, versus 57.61 and 0.6205 for k=3. Despite k=3 providing 2.96× more training examples, the additional lower-ranked samples introduce noise that degrades final performance.

NLI scorer capacity. CED filtering is robust to NLI model capacity: replacing the default scorer (DeBERTa-v3-base, 86M parameters) with DeBERTa-v3-large (304M) yields only +1.10pp Ign-F1 (45.33 vs. 44.23, s42), a difference within the v3-base seed noise of 0.88pp (s42: 44.23, s44: 45.11). Other metrics confirm the pattern: Rel-F1 improves by +1.19pp (46.36 vs. 45.17), Evi-F1 by +0.20pp (81.94 vs. 81.74), while EDCR shifts by −0.39pp (0.6530 vs. 0.6569)—the inconsistent direction across metrics rules out systematic improvement from a larger scorer. This insensitivity supports v3-base as a cost-effective default, as a 3.5× larger scorer does not materially improve CED-based data selection.

Table 5: Full RSFT ablation on DocRED dev set (Qwen3-4B). CED and flat-NLI denote the scoring function used for candidate selection. Best Ign-F1 among evidence-generating methods in **bold**.

Configuration	Ign-F1	Rel-F1	Evi-F1	EDCR
SFT baseline (s42)	39.74	40.49	79.86	0.6982
SFT no-evidence	**47.21**	48.76	–	–
RSFT random k=1	41.14	41.98	79.64	0.6971
RSFT CED k=1 (s42)	44.23	45.17	81.74	0.6569
RSFT CED k=1 (s43)	45.12	46.01	80.63	0.6542
RSFT CED k=1 (s44)	45.11	46.21	80.50	0.6350
RSFT CED k=1 (s45)	44.16	45.14	81.74	0.6550
RSFT CED k=1 (mean)	44.66	45.63	81.15	0.6503
RSFT CED k=3	43.64	44.54	80.88	0.6635
RSFT flat-NLI (s42)	–	45.35	80.43	0.6431
RSFT flat-NLI (s43)	–	45.86	81.58	0.6505
RSFT flat-NLI (mean)	–	45.61	81.01	0.6468
RSFT CED k=1 v3-large (s42)	45.33	46.36	81.94	0.6530

C Scaling Details

Table 6 supplements Table 3 in the main text with per-scale Rel-F1 and Evi-F1 breakdowns. At 1.7B, RSFT improves Rel-F1 from 45.86 to 52.81 (+6.95, multi-seed mean) while achieving Evi-F1 82.63 and EDCR 0.6217. The 4B model shows the same pattern: Rel-F1 increases from 44.39 to 45.80 (+1.41, three-seed mean). For 8B, RSFT brings Rel-F1 from 51.87 to 56.07 (three-seed mean) with

Table 6: Detailed Rel-F1 and Evi-F1 across model scales on DocRED dev set. SFT and RSFT columns show multi-seed means where available (Qwen3 1.7B/4B/8B: 3-seed; 14B: single seed; LLaMA SFT Rel-F1/Evi-F1: s42 only; LLaMA RSFT: 3-seed).

Model	SFT Rel-F1	SFT Evi-F1	RSFT Rel-F1	RSFT Evi-F1	RSFT EDCR
Qwen3-1.7B	45.86	82.85	52.81	82.63	0.6217
Qwen3-4B	44.39	79.61	45.80	80.96	0.6487
Qwen3-8B	51.87	82.94	56.07	81.26	0.6235
Qwen3-14B	56.11	82.92	–	–	–
LLaMA-3.1-8B	55.14	81.51	57.75	82.84	0.5804

Evi-F1 of 81.26 and EDCR 0.6235, confirming tight coupling between extraction and evidence at this scale. At 14B, the SFT baseline already achieves strong Rel-F1 (56.11) and Evi-F1 (82.92).

D Controlled Evaluation Subset

The 1.7B evidence tax comparison (§4.4) is confounded by differential truncation rates. At `max_new_tokens`=1024, the no-evidence model truncates 50.9% of outputs (501/985 documents) versus 31.9% (314/985) for the evidence model. To address this, we adopt a matched@2048 fair evaluation: both configurations are evaluated at `max_new_tokens`=2048, and the tax is computed on the intersection of documents where both produce format-compliant output. This yields 971/985 fair-subset documents, with a tax of -5.63pp (s42), confirming a genuine 1.7B evidence benefit. As additional validation, a stricter controlled subset at @1024 (excluding truncated or repetitive outputs; 442/985 documents) yields -6.2pp, with the stronger benefit attributable to the more aggressive filtering.

Table 7 provides a mechanistic breakdown across three scales (evaluated at @1024). Output duplication rate (fraction of documents containing $\geq$3 repeated triples or unique/total < 0.5) declines monotonically with scale for both training configurations, but the gap between no-evidence and with-evidence models narrows from 27.5pp at 1.7B to 3.7pp at 8B. This pattern supports the implicit regularization hypothesis: evidence generation forces smaller models to attend to supporting context, reducing degenerate repetition. At 4B, duplication is already moderate (31.1% with evidence), so the regularization benefit diminishes; under matched@2048 evaluation, the 4B training-time tax is approximately neutral (+0.25$\pm$0.47pp, three-seed mean). The truncation asymmetry reverses across scales: at 1.7B the no-evidence model truncates 19pp more often; at 8B the evidence model truncates 9pp more often due to the longer output format.

Table 7: Mechanistic breakdown of output quality across scales. Duplication rate: fraction of documents with degenerate outputs. Truncation Δ: no-evidence rate minus with-evidence rate (positive means no-evidence truncates more).

	Duplication Rate			Truncation	
Scale	**w/ evi**	**no-evi**	**Gap**	Δ	**Pattern**
1.7B	33.0	60.5	27.5	+19	no-evi worse
4B	31.1	43.2	12.1	+2	neutral
8B	29.2	32.9	3.7	-9	evi worse

E Training-Time Evidence Tax

The evidence tax can be measured in two ways. The *training-time* tax compares separately trained models (SFT with evidence vs. SFT without evidence in the training output format). The *inference-time* tax (reported in Table 3) evaluates the same RSFT model with and without evidence at inference time.

Naïve comparisons at `max_new_tokens`=1024 are confounded by differential truncation rates (e.g., 50.9% vs. 31.9% at 1.7B). To control for this, we adopt a *matched@2048 fair evaluation*: both configurations are evaluated at `max_new_tokens`=2048, and the tax is computed on the intersection of documents where both models produce format-compliant output. Table 8 reports the matched@2048 training-time tax across Qwen3 scales.

The tax reveals a within-family capacity threshold: evidence generation benefits 1.7B (-5.63pp on 971/985 fair-subset documents) but is approximately neutral at $\geq$4B (+0.25$\pm$0.47pp at 4B across three seeds, +0.04pp at 8B, +0.03pp at 14B). At 1.7B, s42 is the strongest SFT seed (47.86 Ign-F1, +3.22pp above the three-seed mean of 44.64), so the benefit may be amplified relative to the population mean. LLaMA-3.1-8B SFT training-time tax is consistently negative but with high magnitude variance (-1.53 to -11.20pp, mean -5.19 ± 5.25pp under matched@2048) and omitted from the table due to the s43 outlier (16.3% no-evidence format failure inflates its tax); cross-architecture patterns are addressed in §I.

F Why RSFT Avoids Decoupling

The structural difference between selection and optimization explains why RSFT does not exhibit the decoupling observed in gradient-based methods.

Table 8: Training-time evidence tax under matched@2048 fair evaluation (DocRED dev Ign-F1). Both configurations evaluated at `max_new_tokens`=2048; tax computed on format-compliant intersection. Tax = no-evidence − with-evidence; positive = cost.

Model	Fair w/ evi	Fair no-evi	Evi Tax	S
Qwen3-1.7B	48.38	42.75	−5.63	
Qwen3-4B	–	–	$+0.25 \pm 0.47^{(3)}$	
Qwen3-8B	50.15	50.19	+0.04	
Qwen3-14B	55.07	55.10	+0.03	

1.7B and 8B are s42; 4B reports three-seed mean ± std (per-seed range: −0.19 to +0.73pp); 14B evaluated at @1024 with 0% truncation for both configurations (†fair subset equals full dev set). Subset: documents in format-compliant intersection out of 985.

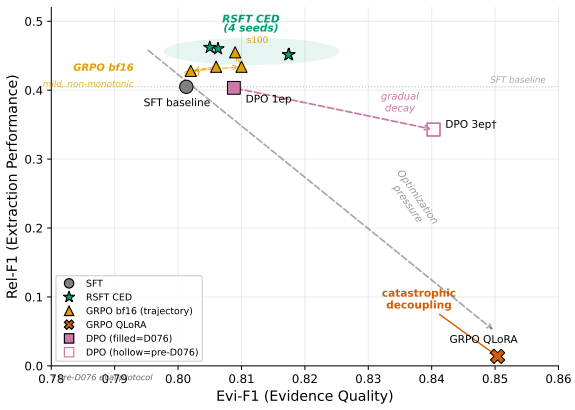


Figure 1: Decoupling spectrum visualization. Each method is plotted by Rel-F1 (x-axis) and Evi-F1 (y-axis). Arrows indicate training progression for DPO (1ep→3ep) and GRPO bf16 trajectory. Ideal methods occupy the upper-right quadrant; decoupled methods drift toward the upper-left.

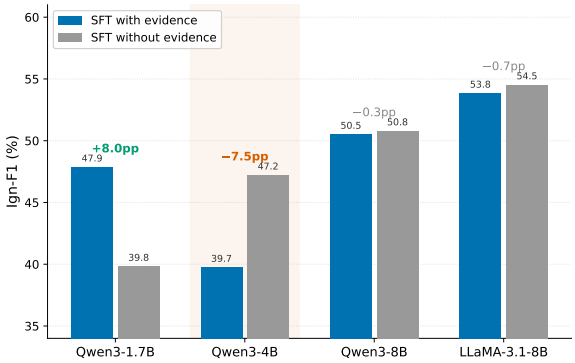


Figure 2: Evidence tax across model scales. Bar heights show Ign-F1 for each training configuration. The gap between no-evidence and SFT bars reverses direction between 1.7B and 4B, revealing the scale-dependent nature of the evidence tax. RSFT consistently narrows the gap toward the no-evidence upper bound.

GRPO and DPO update model parameters through gradients that optimize a reward signal. When two sub-tasks (extraction and evidence) compete for gradient capacity, the optimizer can discover and exploit the trade-off between them, improving one at the expense of the other. RSFT operates differently: it selects complete outputs and trains on them with standard supervised loss. Because the selection is discrete (accept or reject a full response), there is no gradient pathway through which the model can learn to sacrifice extraction for evidence quality. Only outputs where both extraction and evidence are jointly strong survive the CED filter. This property is not specific to RSFT's design but is a structural consequence of discrete selection over complete outputs: any selection-based method that filters on a joint quality criterion will inherit the same coupling guarantee. The implication for practitioners is direct: when optimizing for multiple coupled generation objectives, selection-based methods preserve inter-objective balance that gradient-based methods disrupt.

G Decoupling Spectrum Visualization

H Evidence Tax Visualization

I Full Generalization Results

Cross-family transfer. LLaMA-3.1-8B confirms the scaling patterns observed with Qwen3. Its SFT baseline achieves a three-seed mean of 51.09 Ign-F1 (std 2.92pp; s42: 53.84, s43: 51.42, s44: 48.02). Under matched@2048 fair evaluation, the training-time tax is consistently negative across all three seeds (s42: −1.53, s43: −11.20, s44: −2.83; mean -5.19 ± 5.25pp), indicating that evidence generation consistently benefits LLaMA-3.1-8B. This contrasts with Qwen3-8B's approximately neutral tax (+0.04pp), suggesting that the evidence tax may be architecture-dependent at the 8B scale; however, the high magnitude variance (driven by s43, whose no-evidence configuration has a 16.3% format failure rate that inflates the tax estimate) precludes precise quantification. RSFT CED improves LLaMA to a three-seed mean of 56.54 Ign-F1 (std 0.64pp; s42: 57.14, s43: 56.62, s44: 55.87), a +5.45 point gain over SFT mean (51.09), with 4.6× variance compression (2.92pp → 0.64pp), consistent with the pattern at the 8B scale (Qwen3-8B: +4.14, 6.5× compression).

Cross-dataset robustness. On Re-DocRED (Tan et al., 2022), the SFT baseline achieves 36.59 Ign-

Table 9: SFT and RSFT performance on Re-DocRED test set (Ign-F1 %). Dev Ign-F1 (RSFT) from Table 3. All test results use s42.

Model	Dev	SFT	RSFT	Δ	Tr. SFT	Tr. RSFT	Evi-F1
Qwen3-1.7B	51.19	25.79	29.33	+3.54	81.8	73.8	23.20
Qwen3-4B	44.82†	22.32	25.01	+2.69	64.6	42.8	18.38
Qwen3-8B	54.33	31.06	33.53	+2.47	82.0	1.8	25.26
Qwen3-14B	58.86	38.86	38.49	−0.37	50.6	13.4	26.88
LLaMA-3.1-8B	56.54	34.20	35.99	+1.79	65.4	56.6	25.09

†4B dev Ign-F1 is three-seed mean from original checkpoint; test results use retrained checkpoint (original unavailable for reproduction; retrained v2 achieves comparable dev performance, 43.88 vs. 44.23). Δ: RSFT − SFT. Evi-F1: RSFT joint (DREEAM protocol). Tr.: truncation rate (%). Dev is RSFT three-seed mean from Table 3.

Table 10: Evidence tax on Re-DocRED test set (Ign-F1 %). Tax = no-evi − with-evi; negative = benefit. All results use s42. Dev tax from Table 8.

Model	With-evi	No-evi	Test Tax	Dev Tax
Qwen3-1.7B	25.79	20.93	−4.86	−5.63
Qwen3-4B	22.32	22.19	−0.13	+0.25
Qwen3-8B	31.06	29.35	−1.71	+0.04
Qwen3-14B	38.86	38.54	−0.32	+0.03
LLaMA-3.1-8B	34.20	29.26	−4.94	−5.19

Table 11: RSFT inference-time evidence tax on Re-DocRED test set (Ign-F1 %). Tax = no-evi − with-evi; positive = cost. All results use s42.

Model	RSFT w/ evi	RSFT no-evi	RSFT Tax
Qwen3-1.7B	29.33	23.90	−5.43‡
Qwen3-4B	25.01	26.52	+1.51
Qwen3-8B	33.53	33.19	−0.34
Qwen3-14B	38.49	39.26	+0.77†
LLaMA-3.1-8B	35.99	33.34	−2.65

‡1.7B no-evi truncation rate is 94.8% (vs. 73.8% with evi), which may amplify the tax magnitude. †14B shows a dev-test reversal (dev tax = −4.54pp); the test magnitude (+0.77pp) is small and may reflect noise or evaluation protocol differences.

F1 (Evi-F1 81.70, EDCR 0.6674). RSFT CED improves this to 41.45 Ign-F1 (Evi-F1 81.82, EDCR 0.6210), a +4.86pp gain that closely matches the +5.08pp improvement on DocRED (RSFT three-seed mean minus SFT s42). This consistency across datasets with different annotation protocols confirms that RSFT's benefit generalizes beyond training-set-specific artifacts. Evidence quality remains stable across datasets (Evi-F1 81.82 vs. 81.15 on DocRED), and the EDCR reduction (-0.046) mirrors the DocRED pattern, further supporting the robustness of CED-based selection.

Re-DocRED test set evaluation. Table 9 reports SFT and RSFT performance on the Re-DocRED test set. We observe a consistent performance gap between dev and test sets, with test Ign-F1 approximately 20 percentage points lower (RSFT: 1.7B 51.2→29.3, 4B 44.8→25.0, 8B 54.3→33.5, 14B 58.9→38.5, LLaMA-8B 56.5→36.0; SFT 1.7B: 44.6→25.8). This gap is attributable to annotation density: the test set contains 34.7 gold triples per document versus 12.3 in the dev set (a 2.82× ratio), imposing a harder recall requirement. The gap magnitude is stable across scales and architectures (range: 18.9–21.9pp), and model ranking is perfectly preserved (Spearman ρ=1.0), indicating that the absolute drop reflects the test annotation scheme rather than a model-specific deficiency. At 14B, the RSFT advantage observed on dev (+3.79pp) reverses on test (Δ=−0.37pp); we note that the SFT model's 50.6% truncation rate implies longer outputs that generate more candidate triples to match the test set's dense annotation scheme.

Test-set evidence tax. Table 10 reports the evidence tax on the Re-DocRED test set, comparing SFT models trained with and without evidence in the output format. Test-set evaluation confirms the directional pattern observed on development data across all five model configurations: the tax is negative (beneficial) at 1.7B (−4.86pp) and LLaMA-8B (−4.94pp), and near-neutral at mid-scales (−0.13pp at 4B, −0.32pp at 14B). The 8B test tax (−1.71pp) shows slight variation from its near-neutral dev counterpart (+0.04pp), though both remain modest relative to the 1.7B and LLaMA benefits.

RSFT test-set evidence tax. Table 11 reports the inference-time evidence tax for RSFT models on the Re-DocRED test set (same RSFT model evaluated with vs. without evidence output). At 1.7B, RSFT preserves a strong evidence benefit (−5.43pp), comparable to the SFT tax (−4.86pp). At 4B, the tax is mildly positive (+1.51pp), the only Qwen3 scale where RSFT introduces a measurable evidence cost; this contrasts with the near-neutral SFT tax at 4B (−0.13pp). Compared to the SFT tax in Table 10, RSFT compresses the tax magnitude at 8B (SFT: −1.71pp → RSFT: −0.34pp, 80% reduction). At 14B, the tax reverses sign relative to dev (dev: −4.54pp → test: +0.77pp); the test magnitude is small and may reflect noise or evaluation protocol differences given the 2.82× annotation density gap between dev and test. LLaMA-3.1-8B shows partial compression from SFT (−4.94pp → −2.65pp).

Table 12: Comparison with encoder-based methods on DocRED dev set. Encoder methods use RoBERTa-large; LLM methods use Qwen3. Evi-F1 for encoder methods uses joint evaluation (all triples in denominator); our joint Evi-F1 follows the same protocol for fair comparison. TP-only Evi-F1 conditions on correctly predicted triples. Rel-F1 is reported in percentage for all methods.

Method	Type	Ign-F1	Rel-F1	Evi-F1 (joint)	Evi-F1 (TP)
ATLOP	Encoder	61.32	63.18	–	–
EIDER	Encoder	62.34	64.27	52.54	–
SAIS	Encoder	62.23	65.17	55.84	–
DREEAM	Encoder	61.71	63.49	54.15	–
Ours (4B RSFT)	LLM	44.82	45.80	36.1	80.96
Ours (8B RSFT)	LLM	54.33	56.07	–	81.26
Ours (14B RSFT)	LLM	58.86	60.09	48.59	83.92

Comparison to supervised methods. Table 12 directly compares encoder-based methods with our LLM-based approach. Encoder methods achieve substantially higher extraction performance (Rel-F1 63–65 vs. our 45–56), reflecting decades of architecture-specific optimization for the classification formulation. However, their evidence identification operates as binary sentence classification, producing Evi-F1 of 52–56 under a joint evaluation protocol. Our TP-only Evi-F1 exceeds 80%, indicating that when the model correctly identifies a triple, it produces higher-quality evidence than encoder methods. The joint Evi-F1 gap (36.1 vs. 54.15) is driven primarily by extraction performance rather than evidence quality: false-positive and false-negative triples dilute the denominator. Closing this gap requires improving extraction capability of LLM-based methods, an orthogonal direction to the evidence faithfulness problem we address.

J GRPO Training Dynamics

K Error Analysis: SFT vs. RSFT

Table 13 decomposes the extraction improvements from RSFT into error categories across three scales. Three mechanisms drive the gains consistently:

Truncation false negatives. SFT models frequently hit the generation length limit, producing incomplete outputs that miss triples. RSFT training on compact, high-quality outputs reduces this truncation rate. The effect is strongest at 1.7B (-841 truncation FN, -22%), where the SFT model's output length is least controlled.

Hallucinated false positives. CED-based filtering removes training examples with unfaithful predictions, reducing the rate of hallucinated triples at inference. LLaMA benefits most (-593 hallucinated FP, -11.8%), suggesting cross-architecture variation in hallucination susceptibility.

Table 13: Error pattern changes from SFT to RSFT across scales. $\Delta\%$ shows relative change. Truncation FN, hallucinated FP, and duplicates all decrease; evidence quality and residual FN difficulty are unchanged.

Error Category	Direction	1.7B	4B	8B / LLaMA
Truncation FN	$\downarrow$	-22%	-16%	-18%
Hallucinated FP	$\downarrow$	-8%	-6%	-12%
Duplicates	$\downarrow$	-20%	-11%	-14%
Evidence (exact)	$\approx$	$<$0.4pp	$<$0.4pp	$<$0.4pp
FN: other	$\uparrow$	+13%	+15%	+14%

Duplicate predictions. RSFT training data, selected for quality, contains fewer repetitive patterns. This propagates to inference, reducing duplicate triples by 11–20% across scales, with 1.7B showing the largest reduction (-20.2%).

Evidence quality unchanged. Despite substantial extraction improvements, evidence exact-match F1 changes by less than 0.4pp at every scale. RSFT selects for joint extraction-evidence quality, but the evidence signal in CED scoring does not translate into measurable evidence improvement. This suggests that evidence grounding requires targeted optimization (e.g., evidence-specific rewards in RL) rather than data selection alone.

Residual difficulty. The proportion of "other" false negatives (triples missed for reasons beyond truncation or formatting) increases by 13–15% in relative terms after RSFT. As easy errors (truncation, duplication) are eliminated, the remaining error distribution shifts toward harder cases: cross-sentence reasoning, implicit relations, and long-tail entity pairs. The relation types with the largest RSFT gains (participant, conflict, league, screenwriter, operator, contains_administrative_territory) involve cross-sentence reasoning or long-tail entities, consistent with RSFT's selection pressure toward diverse, high-quality outputs.

L Per-Relation EDCR Distribution

Computing EDCR for each of the 76 relation types with $\geq$5 evaluation samples (from the RSFT CED s42 model) reveals that per-relation EDCR correlates strongly with per-relation Rel-F1 ($\rho = -0.79$, $n = 76$, $p < 0.001$), confirming its utility as a difficulty indicator. Filtering to types with $\geq$5 predictions strengthens the correlation ($\rho = -0.85$,

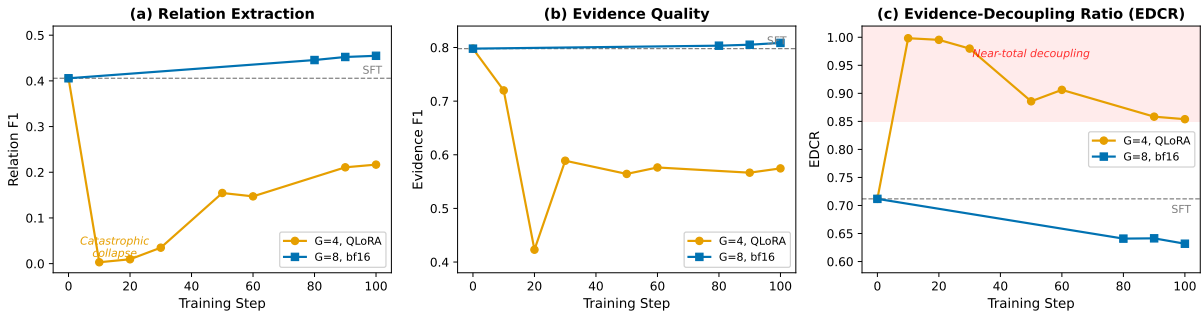


Figure 3: Training dynamics of GRPO under different configurations. (a) Relation F1 across checkpoints: G=4 QLoRA suffers catastrophic collapse at step 10, while G=8 bf16 remains stable. (b) Evidence F1 stays intact for G=8 bf16, confirming the generation mechanism itself does not fail. (c) Evidence-Decoupling Ratio (EDCR) reveals near-total decoupling in G=4 QLoRA (extraction collapses to Rel-F1 1.4), while G=8 bf16 maintains healthy ratios.

$n = 74$, $p < 0.001$). Per-relation EDCR is statistically independent of Evi-F1 ($\rho = -0.06$, $p = 0.58$), and Evi-F1 itself shows no association with Rel-F1 ($\rho = -0.01$, $p = 0.91$). These null correlations establish that evidence quality and extraction ability are fully decoupled across relation types. Extreme cases illustrate the diagnostic power: `date_of_birth` has EDCR $= 0.118$ (highly faithful, with extraction and evidence tightly coupled), while `head_of_government` reaches 0.956 (highly decoupled). This per-relation analysis avoids the shared-component concern present in cross-method EDCR-vs-Rel-F1 correlation, providing independent validation of EDCR's diagnostic utility.

M Qualitative Analysis

We present representative examples illustrating three phenomena: evidence quality differences between SFT and RSFT, GRPO format gaming under catastrophic collapse, and per-relation EDCR diagnostics.

Case 1: Evidence quality. Table 14 compares evidence citations from SFT and RSFT on three documents. SFT tends to cite many sentences indiscriminately, including distractors unrelated to the target triple, while RSFT selects precisely the gold evidence sentences. For *America's Sweetheart*, SFT cites 6 sentences (precision 0.17) whereas RSFT cites only the single correct sentence (precision 1.00). The pattern is consistent: SFT achieves evidence precision of 0.00–0.20 across these examples, while RSFT achieves 1.00 in all three cases.

Case 2: GRPO format gaming. Under catastrophic collapse (GRPO QLoRA G=4), the model abandons the DocRED 96-type relation ontology and generates free-text relation labels. On the document *Something to Believe In*, the model produces 35 predicted triples with relations such as "is a", "honor", and "location", none of which belong to the target ontology. All 35 predictions are incorrect. Similarly, for *Józef Piłsudski Monument*, 28 predictions use non-standard relations with 0 correct matches. The model retains the structural format of triple-with-evidence output but fills it with fabricated relation types. The model produces well-formed evidence for these non-existent relations.

Table 14: Evidence citation comparison between SFT and RSFT. Precision measures the fraction of cited sentences that match gold evidence.

Document	Triple	SFT Prec.	RSFT Prec.	SFT #Cited
Oleg Tinkov	(Oleg Tinkov, country of citizenship, Russia)	0.00	1.00	3
America's Sweetheart	(record label, country, United States)	0.17	1.00	6
Jade (Mortal Kombat)	(Jade, present in work, Mortal Kombat)	0.20	1.00	5

Case 3: EDCR diagnostics. Per-relation EDCR identifies specific relation types where extraction and evidence are misaligned. For `date_of_birth` (EDCR $= 0.118$), the model's evidence consistently matches gold annotations: both extraction and evidence point to the same biographical sentence. In contrast, `notable_work` (EDCR $= 0.941$) exhibits severe decoupling: the model cites sentence [0] (a general introduction) instead of the gold evidence sentence [3] (which contains the actual work attribution). High-EDCR relations tend to involve indirect or multi-hop reasoning where surface-level context misleads the evidence generator.

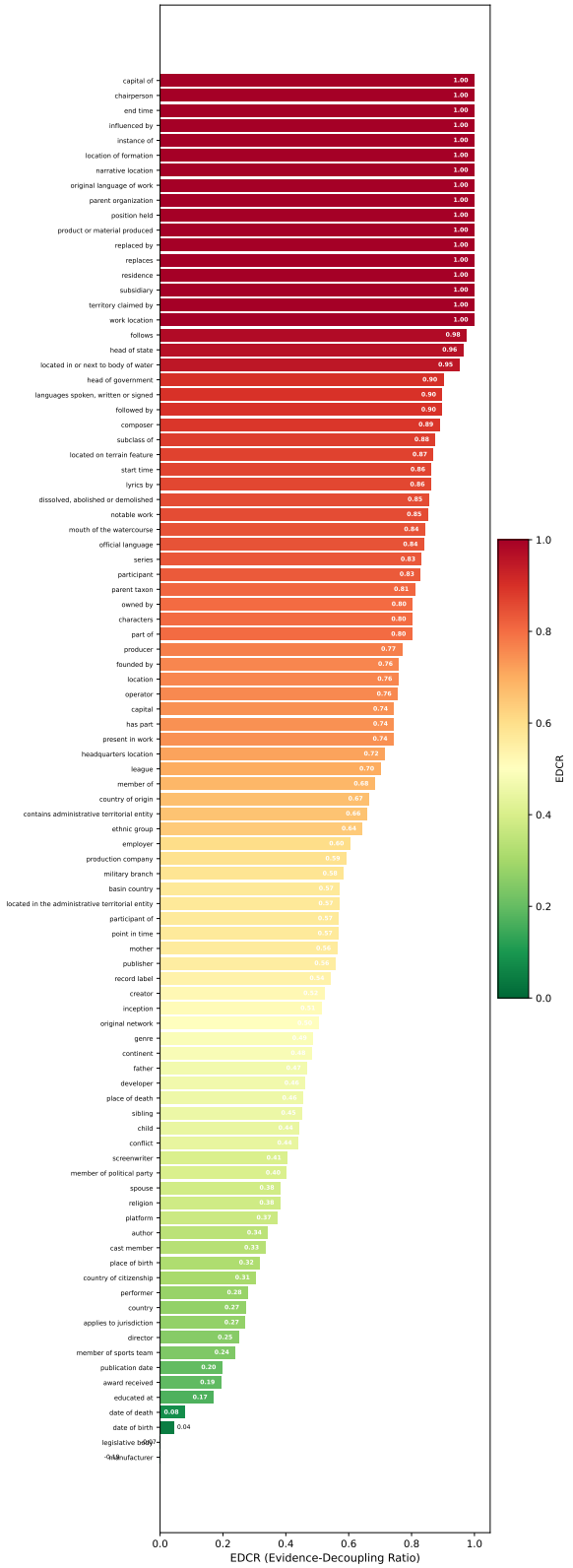


Figure 4: Per-relation EDCR distribution across all 93 relation types observed in the development set (of 96 defined in DocRED). The correlation analysis in the text uses the 76 types with $\geq$5 samples for statistical reliability. Red bars indicate EDCR $\geq$ 0.70 (severe decoupling), green bars indicate EDCR $<$ 0.30 (healthy evidence grounding). Seven types exhibit EDCR = 1.0.